Band-1 Computational Experiment: Dirichlet / Piltz Divisor Problem — Error-Term Exponent Ledger + Empirics

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Dirichlet Divisor Problem (Next Exponent Increment)

Honesty stamp. This note **cites** the best published/arXiv **proven** upper bound on the error exponent α in

$$\Delta(x) = \sum_{n \le x} d(n) - x(\log x + 2\gamma - 1) = O(x^{\alpha+\varepsilon})$$

and reports a **finite-range computational** census of $\Delta(x)$ for integer $x \le 10^7$ plus sparse hyperbola checkpoints to $x = 10^{12}$. It does **not** claim a new proven exponent. Empirical growth rates on a finite interval are **not** proofs; they must not be read as beating Li–Yang or Huxley.

Abstract

Let $d(n)$ be the divisor function and $\Delta(x) = \sum_{n \le x} d(n) - x(\log x + 2\gamma - 1)$. The classical Dirichlet (Piltz $k = 2$) divisor problem asks for the least α such that $\Delta(x) = O(x^{\alpha+\varepsilon})$ for every $\varepsilon > 0$. We ledger historical and current analytic upper bounds, then compute $\Delta(x)$ on this box and report descriptive growth statistics (running-max log–log fit, pointwise $\log|\Delta|/\log x$, sparse large-x ratios).

Literature best proven: Li–Yang (2023), $\alpha \le \alpha^* \approx 0.3144831759741$, improving Huxley (2003) $\alpha \le 131/416 \approx 0.314904$. **Conjecture:** $\alpha = 1/4$. **Box empirics:** descriptive only; **no new proven** α.

Problem statement

Dirichlet divisor problem (error exponent). Determine

$$\alpha_{\text{div}} := \inf\{\alpha : \ \Delta(x) = O(x^{\alpha+\varepsilon}) \ \forall \varepsilon > 0\}.$$

(Piltz generalization: same for d_k, $k = 2$ here.)

This experiment (Band-1, increment). (i) Literature ledger of best known proven α; (ii) exact integer computation of $\Delta(x)$ for $x \le 10^7$; (iii) sparse hyperbola evaluations to 10^{12}; (iv) empirical growth notes, **clearly separated** from proven exponents.

Literature ledger (proven upper bounds on α)

Normalization throughout: $\Delta(x) = O(x^{\alpha+\varepsilon})$.

Year	α bound	Authors	Status
1849	$1/2$	Dirichlet	historical
1903	$1/3 \approx 0.3333$	Voronoï	historical

Year	α bound	Authors	Status
1922	$33/100 = 0.33$	van der Corput	historical
1928	$27/82 \approx 0.3293$	van der Corput	historical
1950–53	$15/46 \approx 0.3261$	Chih / Richert	historical
1969	$12/37 \approx 0.3243$	Kolesnik	historical
1982	$35/108 \approx 0.3241$	Kolesnik	historical
1988	$7/22 \approx 0.3182$	Iwaniec–Mozzochi	published
1993	$23/73 \approx 0.3151$	Huxley	published
2003	$131/416 \approx 0.314904$	**Huxley**	previous record
2023	$\alpha^* \approx 0.3144831759741$	**Li–Yang**	**best known**
—	$1/4$	classical conjecture	**OPEN** (optimal)

Primary citation (best known). Xiaochun Li, Xuerui Yang, *An improvement on Gauss's Circle Problem and Dirichlet's Divisor Problem*, arXiv:2308.14859 (2023). They prove $\Delta(X) = O_\varepsilon(X^{\alpha^*+\varepsilon})$ with $\alpha^* \approx 0.3144831759741$ the unique root in $[0.3, 0.35]$ of their defining equation. Also tabulated in Tao's expdb divisor chapter (Table 16.1).

Previous record. M. N. Huxley, *Exponential sums and lattice points III*, Proc. London Math. Soc. **87** (2003): $\alpha \le 131/416$.

Not elevated. Bourgain–Watt proposed $517/1648 \approx 0.313714$ (arXiv:1709.04340 and related); arXiv comments record **proof gaps**, so Theorems lose status. Careful ledgers (Tao expdb; this note) keep **Li–Yang** as best established / **Huxley** as prior journal anchor. Box does **not** claim 517/1648 as proven.

Lower bound / conjecture. Hardy: $\Delta(x) = \Omega(x^{1/4})$ (with log-powers in $\Omega^{\pm}$ forms); conjecture $\alpha = 1/4$.

Method barrier (context only). Bombieri–Iwaniec / Huxley-type methods face a natural obstacle near $5/16 = 0.3125$; Li–Yang sit just above that barrier. Closing to $1/4$ needs new ideas.

Method (box)

1. **Dense census** $x = 1, \dots, X_{\max} = 10^7$: accumulate $d(n)$ by $O(X \log X)$ sieving (`d[i::i]+=1`), prefix $D(x) = \sum_{n \le x} d(n)$, then $\Delta(x) = D(x) - x(\log x + 2\gamma - 1)$ in float64 for the main term (D exact `int64`).
2. **Sparse checkpoints** via Dirichlet hyperbola $D(x) = 2\sum_{k \le \sqrt{x}} \lfloor x/k \rfloor - \lfloor \sqrt{x} \rfloor^2$ up to $x = 10^{12}$.
3. **Descriptive empirics.** Running max $M(x) = \max_{1 \le n \le x} |\Delta(n)|$; log–log least-squares fit $\log M \approx a + \alpha_{\text{fit}} \log x$ on $x \in [10^3, 10^7]$; pointwise $\alpha_{\text{pt}}(x) = \log|\Delta(x)| / \log x$; ratios $|\Delta|/x^\alpha$ at checkpoints for $\alpha \in \{1/4, \alpha^*, 131/416, 1/3\}$.
4. **Unit tests.** Hyperbola $D(1) = 1$, $D(10) = 27$, $D(100) = 482$; sieve agrees with hyperbola at 10^k through 10^7; $\alpha^* < 131/416$.

Complexity. Dense: $O(X \log X)$ sieve + $O(X)$ prefix; wall $\sim$ 16 s at $X = 10^7$. Sparse: $O(\sqrt{x})$ per checkpoint.

Results (numbers!)

Unit tests

`PASS`.

Literature (cited, not re-proved)

Quantity	Value
Best proven α	$\alpha^* \approx 0.3144831759741$ **(Li–Yang 2023)**
Previous	$131/416 \approx 0.314903846154$ (Huxley 2003)

Quantity	Value
NOT elevated	$517/1648 \approx 0.313714$ (Bourgain–Watt; gaps)
Conjecture	$1/4$
Hardy Ω-lower	$\alpha \geq 1/4$

Box dense census ($x = 1, \ldots, 10^7$)

Quantity	Value
$X_{\max}$	**10000000**
$\max \|\Delta\|$	**296.629331** at $x = 7876458$
$D(10^7)$	162725364
$\Delta(10^7)$	94.192386
Running-max log–log α_{fit}	≈ 0.29489 (descriptive)
$\max_{x \geq 10^3} \alpha_{\text{pt}}(x)$	≈ 0.41593 (spikes; not an upper bound)
Mean α_{pt} on upper half	≈ 0.2082
Wall time	**15.94 s**

Dense checkpoint ratios $|\Delta(x)|/x^\alpha$

x	$\|\Delta\|$	$/x^{1/4}$	$/x^{\alpha^*}$	$/x^{131/416}$
10^2	6.040	1.910	1.419	1.417
10^3	6.813	1.212	0.776	0.774
10^4	20.283	2.028	1.120	1.116
10^5	14.321	0.805	0.383	0.381
10^6	92.112	2.913	1.195	1.188
10^7	94.192	1.675	0.592	0.588

Sparse hyperbola (selected)

x	$\|\Delta\|$	$\|\Delta\|/x^{1/4}$	α_{pt}
10^8	360.62	3.606	0.320
10^9	467.25	2.628	0.297
10^{10}	622.56	1.969	0.279
10^{11}	1023.24	1.820	0.274
10^{12}	3354.39	3.354	0.294

(Peak dense $|Delta| = 296.63$ at $x = 7876458$ gives $|\Delta|/x^{1/4} \approx 5.60$ — slow $x^{1/4}$-normalized envelope growth, consistent with Hardy-type log-power heuristics, **not** a theorem. Full sample: `Delta_of_x.csv`.)

Figure

Empirics vs proven — separation

Claim type	Content	Status
Proven upper	$\alpha \leq \alpha^* \approx 0.314483$	Li–Yang 2023 (cited)
Proven previous	$\alpha \leq 131/416 \approx 0.314904$	Huxley 2003 (cited)

Claim type	Content	Status
Conjecture	$\alpha = 1/4$	OPEN
Box $\alpha_{\text{fit}} \approx 0.295$	descriptive fit of running-max $\|\Delta\|$ on $x \le 10^7$	**NOT a proof**; finite-range only
Box α_{pt}	oscillates; local spikes $> \alpha^*$ mean nothing for lim sup proofs	**NOT a proof**
Sparse $\|\Delta\|/x^{1/4}$	order-1 to few units up to 10^{12}	**illustration only**

Do not claim: any improvement on Li–Yang / Huxley; conjecture settled; that $\alpha_{\text{fit}} < \alpha^*$ implies a new theorem; that Bourgain–Watt is established.

Why Band-1 vs Band-4

- **Band-1 (this note):** ledger the **next cited exponent increment** (Huxley → Li–Yang) and exercise exact / hyperbola divisor-sum error computation + empirical growth diagnostics.
- **Band-4:** a proof that $\alpha = 1/4$ (or any new rigorous upper bound $< \alpha^*$). Still **OPEN**; **not claimed**.

Reproducibility

- Code: `/workspace/experiments/band1/divisor_problem/`
- Run: `python3 divisor_problem.py`
- Artifacts: `results.json`, `literature_ledger.json`, `Delta_of_x.csv`, `SUMMARY.txt`, `error_growth.png` / `.pdf`
- Log: `/workspace/experiments/logs/2026-09-11_divisor_problem.log`

Conclusion

Literature: best known proven exponent is $\alpha \le \alpha^* \approx 0.3144831759741$ **(Li–Yang 2023)**, a small but real increment past Huxley's $131/416 \approx 0.314904$. Conjecture remains $\alpha = 1/4$. Bourgain–Watt $517/1648$ is **not elevated** here (proof gaps).

Box: exact $\Delta(x)$ for integer $x \le 10^7$; running-max descriptive fit $\alpha_{\text{fit}} \approx 0.295$; peak $|\Delta| \approx 297$ at $x = 7876458$; sparse hyperbola through 10^{12} with $|\Delta|/x^{1/4}$ remaining $O(1)$–few. Empirics are compatible with Hardy-scale growth on this range and **do not** constitute a new proven exponent.

Live Band-1 target: larger dense X, refined envelope statistics, ledger hygiene — **without claiming the divisor problem is solved.**

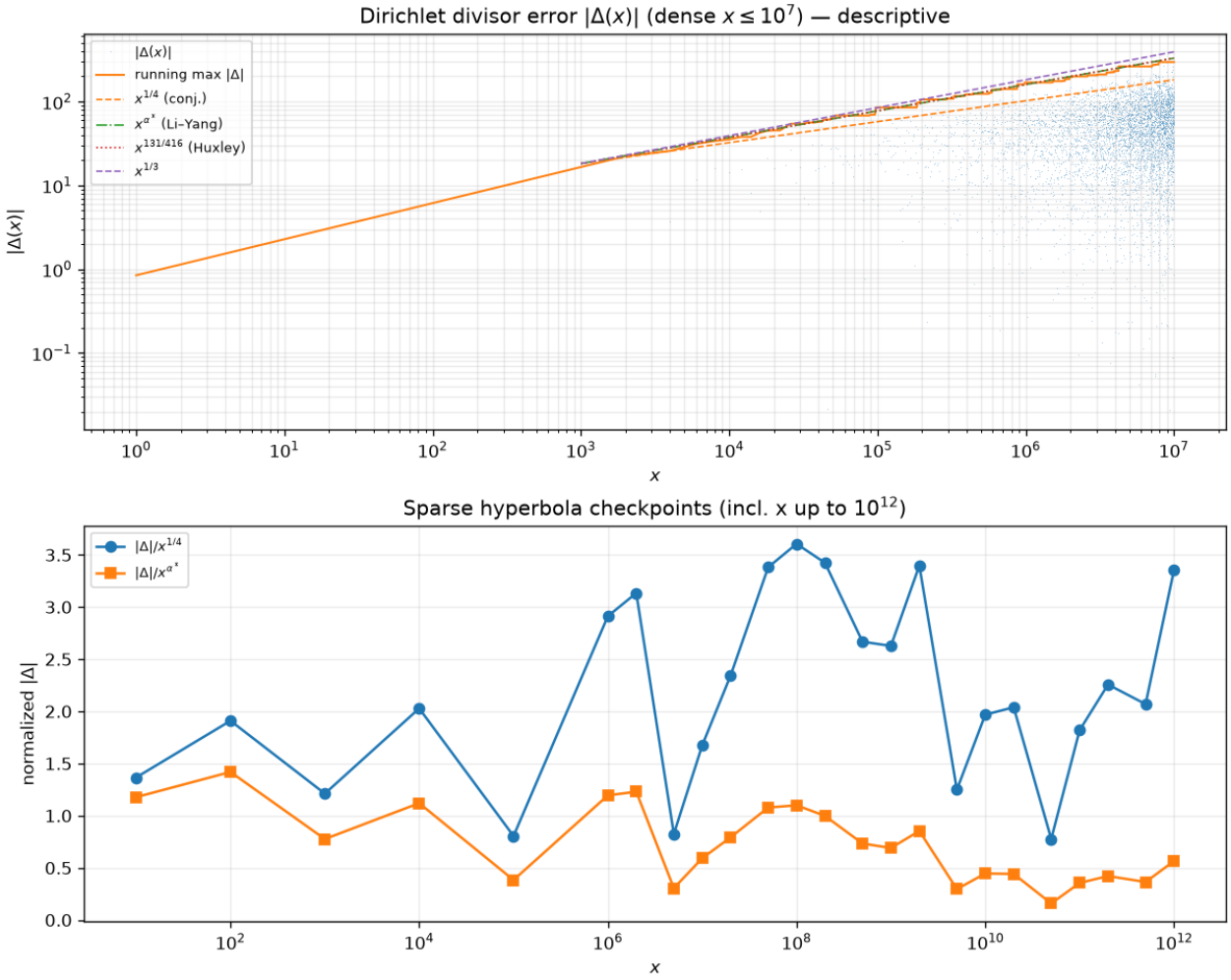


Figure 1: Error growth